affirmoTM tutorial 6: walkthrough

One run followed in detail: patients, costs, QALYs and how the arms behave over time

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1 About this tutorial

This tutorial follows the example inputs through the model: what goes in, what each function produces, and how the two arms behave over time.

- **Inputs:** each one is read with the package's own reader and explored with short `data.table` commands.
- **One run, followed step by step:** a few patients are followed year by year, with their costs and QALYs worked out by hand.
- **Over time:** the cohort is followed across the horizon in both arms.
- **The full model:** `tm_run()` at the end.

It starts from a fresh analysis folder with the example input files (see tutorial 1):

```
library(affirmoTM)
library(data.table)  # for dcast(), melt() and the other table functions used below
library(ggplot2)     # for the plots
```

```
tm_example_inputs()
```

2 The inputs

2.1 Settings

```
settings <- tm_read_settings()
str(settings)
```

```
List of 9
 $ strata               : chr [1:3] "agegrp" "sex" "mm_cluster"
 $ seed                 : int 2026
 $ horizon              : int 10
 $ age_limit_death_state: chr "CurrentDeath"
 $ discount_rate        : num 0.03
 $ n_runs               : int 100
 $ events               :List of 3
  ..$ ischaemic_stroke   : chr [1:5] "AIS" "AIS_AB" "PB_AIS" "AIS_Death" ...
  ..$ haemorrhagic_stroke: chr [1:5] "AHS" "AHS_AB" "PB_AHS" "AHS_Death" ...
  ..$ major_bleed        : chr [1:7] "AB" "AIS_AB" "AHS_AB" "PS_AB" ...
 $ currency             : chr "€"
 $ wtp                  : int [1:3] 20000 30000 50000
```

2.2 Transition matrices

`tm_read_matrices()` reads both matrices and checks them. Each one is a `tm_matrix`; printing it gives an overview:

```
matrices <- tm_read_matrices(settings)
matrices$usual_care
```

```
<tm_matrix> transition matrix (source: user)
  Strata:         agegrp (9), sex (2), mm_cluster (7) -> 126 strata
  Living states:  13 - NoEvent, AB, AHS, AIS, PB, PHS, PIS, AHS_AB, AIS_AB, PB_AHS, PB_AIS, PS_AB, PS_PB
  Death states:   6 - CurrentDeath, AB_Death, AHS_Death, AIS_Death, AHS_AB_Death, AIS_AB_Death
  Years:          the same rows every year
  Draws:          none (point values)
```

The probabilities are in `probs`, one row per stratum and current state (`from`), and one column per next state:

```
usual_care <- matrices$usual_care$probs
iabc <- matrices$iabc$probs
states <- matrices$usual_care$states
death <- matrices$usual_care$death
strata <- settings$strata
age_order <- unique(usual_care$agegrp)[order(as.integer(sub("-.*", "",
 ↪  unique(usual_care$agegrp))))]  # for plots
arm_colours <- c(usual_care = "grey35", iabc = "#2a78d6")
dim(usual_care)
```

```
[1] 2394   23
```

```
usual_care[1:3, 1:10]
```

```
   agegrp    sex  mm_cluster    from   NoEvent          AB         AHS         AIS
   <char> <char>      <char>  <char>     <num>       <num>       <num>       <num>
1:  70-74    men Unspecified NoEvent 0.9693759 0.004641617 0.002450905 0.006487908
2:  75-79    men Unspecified NoEvent 0.9571437 0.005880312 0.002995982 0.008038839
3:  65-69    men Unspecified NoEvent 0.9753106 0.003494647 0.001866399 0.005250628
      PB   PHS
   <num> <num>
1:     0     0
2:     0     0
3:     0     0
```

Every row sums to 1, in both matrices:

```
usual_care[, .(min = min(rowSums(.SD)), max = max(rowSums(.SD))), .SDcols = states]
```

```
     min   max
   <num> <num>
1:     1     1
```

```
iabc[, .(min = min(rowSums(.SD)), max = max(rowSums(.SD))), .SDcols = states]
```

```
     min   max
   <num> <num>
1:     1     1
```

The matrices are easier to explore in long form, one row per move, with both arms side by side:

```
long_usual_care <- melt(usual_care, id.vars = c(strata, "from"), variable.name = "to",
 ↪  value.name = "p",
                         variable.factor = FALSE)
long_iabc <- melt(iabc, id.vars = c(strata, "from"), variable.name = "to", value.name = "p",
 ↪  variable.factor = FALSE)
moves <- long_usual_care[long_iabc, on = c(strata, "from", "to"),
                         .(agegrp, sex, mm_cluster, from, to, usual_care = p, iabc = i.p)]
moves
```

```
        agegrp    sex  mm_cluster         from           to usual_care      iabc
        <char> <char>      <char>       <char>       <char>      <num>     <num>
    1:   70-74    men Unspecified      NoEvent      NoEvent  0.9693759 0.9707387
    2:   75-79    men Unspecified      NoEvent      NoEvent  0.9571437 0.9588437
    3:   65-69    men Unspecified      NoEvent      NoEvent  0.9753106 0.9763731
    4:   80-84    men Unspecified      NoEvent      NoEvent  0.9348558 0.9370792
    5:   85-89    men Unspecified      NoEvent      NoEvent  0.8995875 0.9022077
   ---
45482: 105-109  women   Metabolic     AB_Death AIS_AB_Death  0.0000000 0.0000000
45483: 105-109  women   Metabolic    AHS_Death AIS_AB_Death  0.0000000 0.0000000
45484: 105-109  women   Metabolic    AIS_Death AIS_AB_Death  0.0000000 0.0000000
45485: 105-109  women   Metabolic AHS_AB_Death AIS_AB_Death  0.0000000 0.0000000
45486: 105-109  women   Metabolic AIS_AB_Death AIS_AB_Death  1.0000000 1.0000000
```

A death state's rows always go back to the same state. In every stratum, the only move out of a death state is to itself, with probability 1:

```
moves[from %in% death & usual_care > 0, .(rows = .N, probability = unique(usual_care)), by =
 ↪  .(from, to)]
```

```
          from           to  rows probability
        <char>       <char> <int>       <num>
1: CurrentDeath CurrentDeath   126           1
2:     AB_Death     AB_Death   126           1
3:    AHS_Death    AHS_Death   126           1
4:    AIS_Death    AIS_Death   126           1
5: AHS_AB_Death AHS_AB_Death   126           1
6: AIS_AB_Death AIS_AB_Death   126           1
```

The moves that are possible from each state, i.e. with a probability above 0 in at least one stratum:

```
knitr::kable(moves[usual_care > 0, .(possible_next_states = paste(unique(to), collapse = ",
 ↪  ")), by = from])
```

from	possible_next_states
NoEvent	NoEvent, AB, AHS, AIS, AHS_AB, AIS_AB, CurrentDeath, AB_Death, AHS_Death, AIS_Death, AHS_AB_Death, AIS_AB_Death
PB	AB, PB, PB_AHS, PB_AIS, CurrentDeath, AB_Death
PHS	AHS, PHS, PS_AB, CurrentDeath
PIS	AIS, PIS, PS_AB, CurrentDeath, AHS_Death, AIS_Death
AB	PB, PB_AHS, PB_AIS, CurrentDeath, AB_Death, AHS_Death, AIS_Death, AIS_AB_Death
AHS	PHS, AHS_AB, PS_AB, CurrentDeath, AHS_Death, AIS_Death
AIS	PIS, AIS_AB, PS_AB, CurrentDeath, AB_Death, AHS_Death, AIS_Death
PS_PB	PB_AHS, PB_AIS, PS_AB, PS_PB, CurrentDeath
AHS_AB	PS_PB, CurrentDeath
AIS_AB	PS_PB, CurrentDeath, AB_Death
PB_AIS	PS_PB, CurrentDeath
PS_AB	PS_PB, CurrentDeath
PB_AHS	PS_PB, CurrentDeath
CurrentDeath	CurrentDeath
AB_Death	AB_Death
AHS_Death	AHS_Death
AIS_Death	AIS_Death
AHS_AB_Death	AHS_AB_Death
AIS_AB_Death	AIS_AB_Death

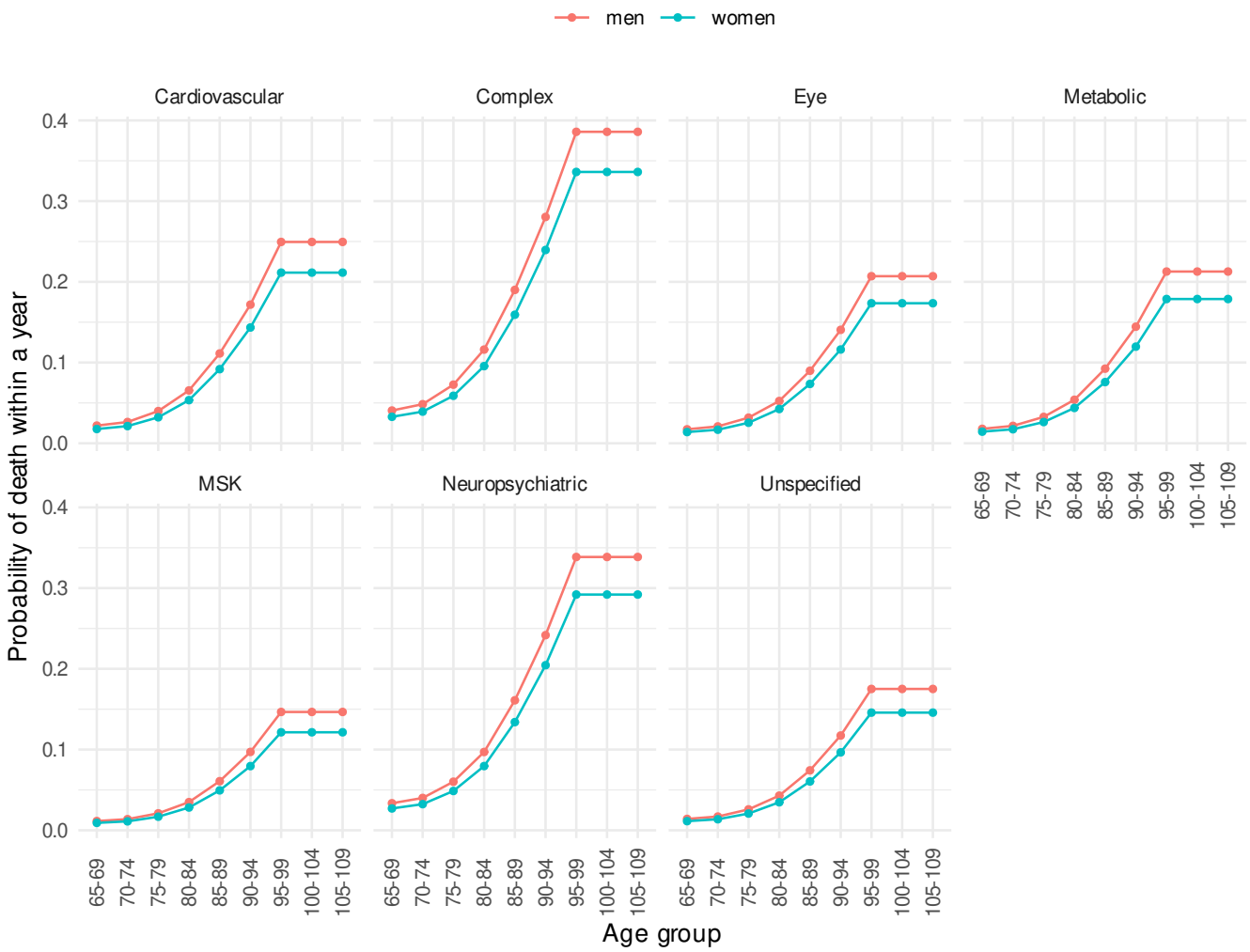
The same patient under each arm: a man aged 75–79 in the Unspecified cluster, without an event this year. Each number is the probability of being in that state next year:

```
moves[agegrp == "75-79" & sex == "men" & mm_cluster == "Unspecified" & from == "NoEvent" &
 ↪  usual_care > 0,
     .(to, usual_care, iabc, difference = iabc - usual_care)]
```

```
            to    usual_care          iabc     difference
          <char>         <num>         <num>          <num>
 1:      NoEvent  9.571437e-01  9.588437e-01   1.699973e-03
 2:           AB  5.880312e-03  5.292281e-03  -5.880312e-04
 3:          AHS  2.995982e-03  2.696384e-03  -2.995982e-04
 4:          AIS  8.038839e-03  7.234955e-03  -8.038839e-04
 5:       AHS_AB 7.894549e-100 7.105094e-100 -7.894549e-101
 6:       AIS_AB  8.459520e-05  7.613568e-05  -8.459520e-06
 7: CurrentDeath  2.364729e-02  2.364729e-02   0.000000e+00
 8:     AB_Death  4.052030e-04  4.052030e-04   0.000000e+00
 9:    AHS_Death  7.969846e-04  7.969846e-04   0.000000e+00
10:    AIS_Death  9.614472e-04  9.614472e-04   0.000000e+00
11: AHS_AB_Death 1.183016e-164 1.183016e-164   0.000000e+00
12: AIS_AB_Death  4.563169e-05  4.563169e-05   0.000000e+00
```

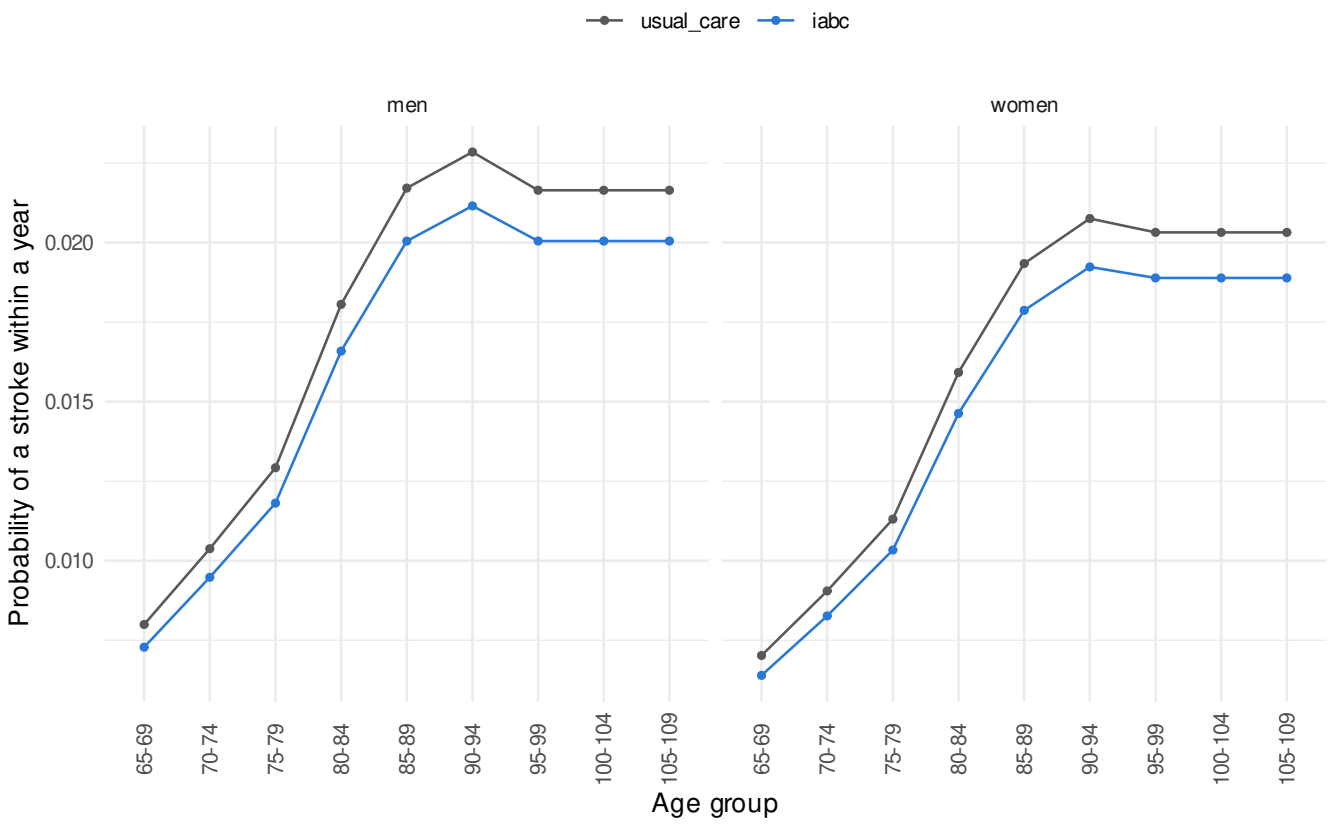
The yearly probability of death for patients without an event, by age group, sex and cluster:

```
death_by_age <- moves[from == "NoEvent" & to %in% death, .(death = sum(usual_care)), by =
 ↪  .(agegrp, sex, mm_cluster)]
ggplot(death_by_age, aes(factor(agegrp, levels = age_order), death, colour = sex, group = sex))
 ↪  +
  geom_line() + geom_point(size = 1) +
  facet_wrap(~mm_cluster, ncol = 4) +
  labs(x = "Age group", y = "Probability of death within a year", colour = NULL) +
  theme_minimal(base_size = 10) + theme(axis.text.x = element_text(angle = 90, vjust = 0.5),
   ↪  legend.position = "top")
```



The yearly probability of a stroke for patients without an event, under each arm, for the Unspecified cluster:

```
stroke_states <- unique(unlist(settings$events[c("ischaemic_stroke", "haemorrhagic_stroke")]))
stroke_by_age <- moves[from == "NoEvent" & to %in% stroke_states & mm_cluster == "Unspecified",
                       .(usual_care = sum(usual_care), iabc = sum(iabc)), by = .(agegrp, sex)]
ggplot(melt(stroke_by_age, id.vars = c("agegrp", "sex"), variable.name = "arm", value.name =
 ↪  "stroke"),
       aes(factor(agegrp, levels = age_order), stroke, colour = arm, group = arm)) +
  geom_line() + geom_point(size = 1) + facet_wrap(~sex) + scale_colour_manual(values =
   ↪  arm_colours, breaks = names(arm_colours)) +
  labs(x = "Age group", y = "Probability of a stroke within a year", colour = NULL) +
  theme_minimal(base_size = 10) + theme(axis.text.x = element_text(angle = 90, vjust = 0.5),
   ↪  legend.position = "top")
```



2.3 Cohort

`tm_read_cohort()` reads the patients and works out each patient's age group from their age:

```
cohort <- tm_read_cohort(settings, matrices)
cohort
```

```
         id entry_cycle   age agegrp    sex       mm_cluster
      <int>       <int> <int> <char> <char>           <char>
   1:     1           0    65  65-69    men Neuropsychiatric
   2:     2           0    65  65-69    men Neuropsychiatric
   3:     3           0    65  65-69  women          Complex
   4:     4           0    65  65-69  women              Eye
   5:     5           0    65  65-69  women   Cardiovascular
  ---
4996:  4996           9    94  90-94  women      Unspecified
4997:  4997           9    95  95-99  women          Complex
4998:  4998           9    96  95-99  women              Eye
4999:  4999           9    97  95-99  women          Complex
5000:  5000           9    99  95-99  women              MSK
```

Patients enter in different cycles. Cycle 0 (2013) holds everyone already living with AF when the data start, and later cycles hold new patients:

```
cohort[, .(patients = .N, mean_age = round(mean(age), 1)), keyby = entry_cycle]
```

```
Key: <entry_cycle>
    entry_cycle patients mean_age
          <int>    <int>    <num>
```

```
 1:           0     2664     78.6
 2:           1      245     74.0
 3:           2      272     75.1
 4:           3      236     74.1
 5:           4      240     74.1
 6:           5      272     73.3
 7:           6      249     75.3
 8:           7      251     74.5
 9:           8      260     74.5
10:           9      311     75.1
```

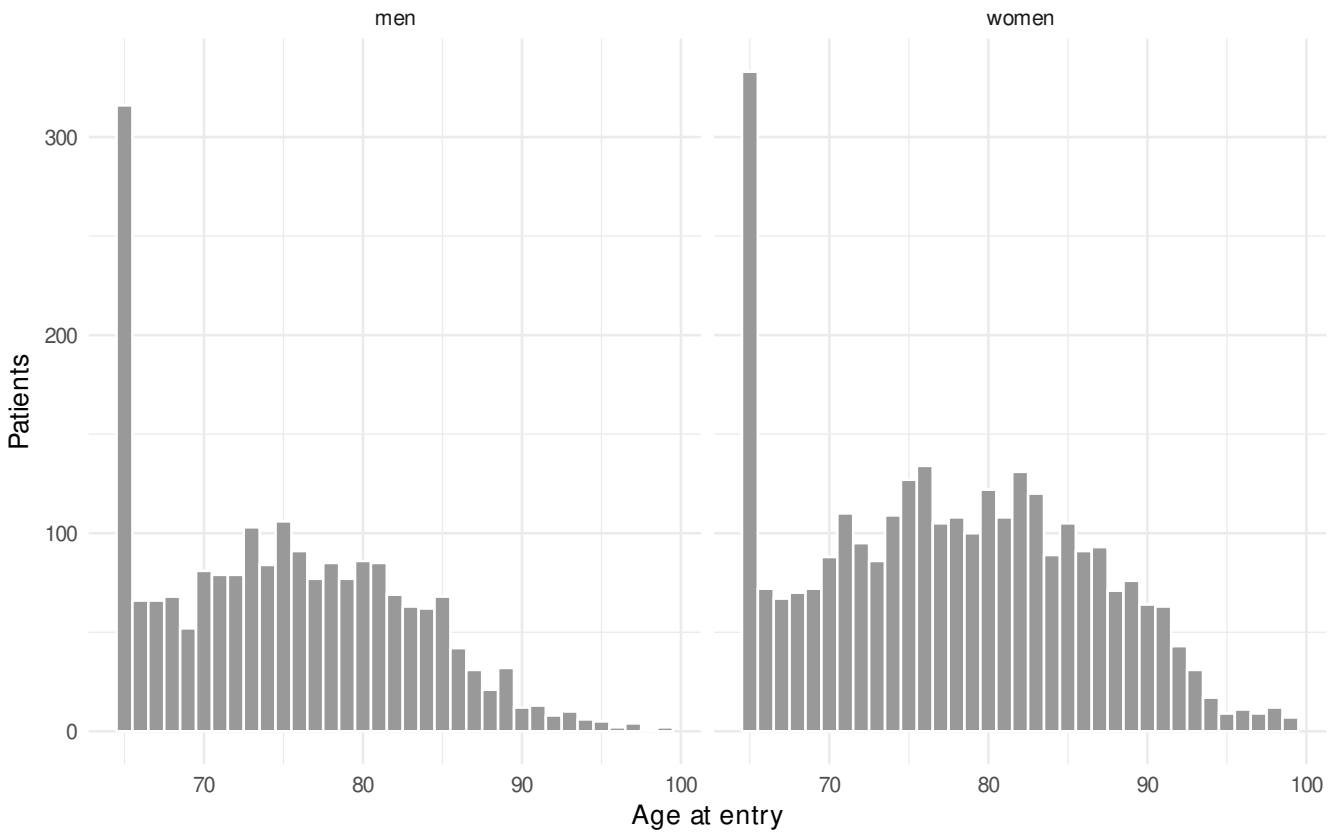
```
cohort[, .N, keyby = sex]
```

```
Key: <sex>
      sex     N
   <char> <int>
1:    men  2052
2:  women  2948
```

```
cohort[, .N, keyby = mm_cluster]
```

```
Key: <mm_cluster>
         mm_cluster     N
              <char> <int>
1:   Cardiovascular  1720
2:          Complex   362
3:              Eye   603
4:              MSK   881
5:        Metabolic   345
6: Neuropsychiatric   721
7:      Unspecified   368
```

```
ggplot(cohort, aes(age)) +
  geom_histogram(binwidth = 1, fill = "grey60", colour = "white") +
  facet_wrap(~sex) +
  labs(x = "Age at entry", y = "Patients") +
  theme_minimal(base_size = 10)
```



2.4 Entry states

`tm_read_entry_states()` reads the probability of each state in a patient's entry year, one row per stratum. Every row sums to 1:

```
entry_states <- tm_read_entry_states(settings, matrices)
entry_cols <- setdiff(names(entry_states), strata)
entry_states[, .(rows = .N, min = min(rowSums(.SD)), max = max(rowSums(.SD))), .SDcols =
 ↪  entry_cols]
```

```
    rows   min   max
   <int> <num> <num>
1:    98     1     1
```

```
entry_states[agegrp == "75-79" & sex == "women" & mm_cluster == "Cardiovascular"]
```

```
   agegrp    sex     mm_cluster   NoEvent    AB         AHS        AIS          PB
   <char> <char>         <char>     <num> <num>       <num>      <num>       <num>
1:  75-79  women Cardiovascular 0.9011299     0 0.004708098 0.01506591 0.005649718
          PHS        PIS AHS_AB AIS_AB PB_AHS PB_AIS PS_AB PS_PB CurrentDeath
        <num>      <num>  <num>  <num>  <num>  <num> <num> <num>        <num>
1: 0.003766478 0.01506591      0      0      0      0     0     0   0.05461394
```

2.5 Costs, utilities and the intervention cost

```r
costs <- tm_read_costs(settings, matrices)
utilities <- tm_read_utilities(settings, matrices)
intervention <- tm_read_intervention_cost(settings)
costs
```

```
          state usual_care  iabc
         <char>      <num> <num>
 1:      NoEvent       2000  2000
 2:           AB       8000  8000
 3:          AHS      20000 20000
 4:          AIS      15000 15000
 5:           PB       3000  3000
 6:          PHS       5000  5000
 7:          PIS       4000  4000
 8:       AHS_AB      27000 27000
 9:       AIS_AB      22000 22000
10:       PB_AHS      22000 22000
11:       PB_AIS      17000 17000
12:        PS_AB      12000 12000
13:        PS_PB       6000  6000
14: CurrentDeath       3000  3000
15:     AB_Death       7000  7000
16:    AHS_Death      15000 15000
17:    AIS_Death      12000 12000
18: AHS_AB_Death      20000 20000
19: AIS_AB_Death      18000 18000
```

```r
utilities
```

```
          state usual_care  iabc
         <char>      <num> <num>
 1:      NoEvent       0.75  0.75
 2:           AB       0.65  0.65
 3:          AHS       0.50  0.50
 4:          AIS       0.55  0.55
 5:           PB       0.70  0.70
 6:          PHS       0.60  0.60
 7:          PIS       0.65  0.65
 8:       AHS_AB       0.40  0.40
 9:       AIS_AB       0.45  0.45
10:       PB_AHS       0.47  0.47
11:       PB_AIS       0.52  0.52
12:        PS_AB       0.55  0.55
13:        PS_PB       0.60  0.60
14: CurrentDeath       0.00  0.00
15:     AB_Death       0.00  0.00
16:    AHS_Death       0.00  0.00
17:    AIS_Death       0.00  0.00
18: AHS_AB_Death       0.00  0.00
19: AIS_AB_Death       0.00  0.00
```

```r
intervention
```

```
    year  cost
   <int> <num>
1:     1    50
2:     2     0
3:     3     0
4:     4     0
5:     5     0
6:     6     0
```

```
7:     7     0
8:     8     0
9:     9     0
```

- **Costs:** the cost of a year spent in each state, per arm.
- **Utilities:** the QALY weight of a year in each state, per arm. Death states count 0.
- **Intervention cost:** the cost of delivering iABC to one patient in each year after entry. It is charged to iABC patients who are alive in that year.

3 One run, step by step

`tm_run()` repeats these steps once per run. Here they are done one at a time for the first run, which uses the seed from `settings.yaml`:

```
seed <- settings$seed
seed
```

```
[1] 2026
```

3.1 Entry states

`tm_assign_entry_states()` draws each patient's state in their entry year from their row of the entry-state table. Both arms start from these states.

```
patients <- tm_assign_entry_states(cohort, entry_states, settings, seed = seed)
patients[, .N, keyby = entry_state][order(-N)]
```

```
     entry_state     N
          <char> <int>
 1:      NoEvent  4318
 2: CurrentDeath   420
 3:          PIS    89
 4:          AIS    64
 5:           PB    43
 6:          PHS    28
 7:           AB    17
 8:          AHS    16
 9:       PB_AIS     2
10:        PS_AB     2
11:       AHS_AB     1
```

3.2 The simulation

`tm_simulate()` moves every patient through both arms one year at a time. In each later year, it draws the patient's next state from their matrix row: their arm, sex and cluster, their age group in the previous year, and their previous state. The result has one row per patient, arm and year, from the entry year up to and including the year of death:

```
sim <- tm_simulate(patients, matrices, settings, seed = seed)
setorder(sim, arm, id, cycle)
sim
```

```
          id        arm cycle   age agegrp    sex       mm_cluster        state
       <int>     <char> <int> <int> <char> <char>           <char>       <char>
    1:     1       iabc     0    65  65-69    men Neuropsychiatric      NoEvent
    2:     1       iabc     1    66  65-69    men Neuropsychiatric      NoEvent
    3:     1       iabc     2    67  65-69    men Neuropsychiatric      NoEvent
    4:     1       iabc     3    68  65-69    men Neuropsychiatric      NoEvent
    5:     1       iabc     4    69  65-69    men Neuropsychiatric      NoEvent
   ---
59551:  4996 usual_care     9    94  90-94  women      Unspecified      NoEvent
59552:  4997 usual_care     9    95  95-99  women          Complex      NoEvent
59553:  4998 usual_care     9    96  95-99  women              Eye CurrentDeath
59554:  4999 usual_care     9    97  95-99  women          Complex CurrentDeath
59555:  5000 usual_care     9    99  95-99  women              MSK CurrentDeath
```

```
sim[, .(patients = uniqueN(id), patient_years = .N, deaths = sum(state %in% death)), by = arm]
```

```
         arm patients patient_years deaths
      <char>    <int>         <int>  <int>
1:       iabc     5000         29782   1845
2: usual_care     5000         29773   1846
```

3.3 Following five patients

Each patient's path, written as one line per arm, makes it easy to pick patients with different stories:

```
paths <- sim[, .(path = paste(state, collapse = " > "), entered = min(cycle), last_state =
 ↪  last(state)), by = .(arm, id)]
paths <- dcast(paths, id + entered ~ arm, value.var = c("path", "last_state"))
head(paths[, .(id, entered, path_usual_care)])
```

```
Key: <id, entered>
      id entered
   <int>   <int>
1:     1       0
2:     2       0
3:     3       0
4:     4       0
5:     5       0
6:     6       0
                                                                                path_usual_care
                                                                                         <char>
1: NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent
2:                     NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > AIS > PIS > PIS > PIS > PIS
3: NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent
4:                          NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > CurrentDeath
5: NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent
6: NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent
```

The five patients, chosen from these paths:

```
event_free <- paths[entered == 0 & path_usual_care == paste(rep("NoEvent", 10), collapse = " >
 ↪  "), id][1]
stroke <- paths[entered == 0 & grepl("NoEvent > AIS > PIS", path_usual_care), id][1]
died <- paths[entered == 0 & last_state_usual_care == "CurrentDeath" & grepl("^NoEvent >
 ↪  NoEvent > NoEvent", path_usual_care), id][1]
late <- paths[entered == 7 & path_usual_care == path_iabc, id][1]
differs <- paths[grepl("AIS", path_usual_care) & !grepl("AIS", path_iabc), id][1]
c(event_free = event_free, stroke = stroke, died = died, late = late, differs = differs)
```

```
event_free     stroke       died       late    differs
         1          2          4       4179        476
```

Each patient's years in both arms, side by side:

A patient who stays event-free for the whole horizon:

```
dcast(sim[id == event_free], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
    cycle   age agegrp    iabc usual_care
    <int> <int> <char>  <char>     <char>
 1:     0    65  65-69 NoEvent    NoEvent
 2:     1    66  65-69 NoEvent    NoEvent
 3:     2    67  65-69 NoEvent    NoEvent
 4:     3    68  65-69 NoEvent    NoEvent
 5:     4    69  65-69 NoEvent    NoEvent
 6:     5    70  70-74 NoEvent    NoEvent
 7:     6    71  70-74 NoEvent    NoEvent
 8:     7    72  70-74 NoEvent    NoEvent
 9:     8    73  70-74 NoEvent    NoEvent
10:     9    74  70-74 NoEvent    NoEvent
```

A patient with an ischaemic stroke: acute (`AIS`) in the year it happens, then past (`PIS`) in the years after:

```
dcast(sim[id == stroke], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
    cycle   age agegrp    iabc usual_care
    <int> <int> <char>  <char>     <char>
 1:     0    65  65-69 NoEvent    NoEvent
 2:     1    66  65-69 NoEvent    NoEvent
 3:     2    67  65-69 NoEvent    NoEvent
 4:     3    68  65-69 NoEvent    NoEvent
 5:     4    69  65-69 NoEvent    NoEvent
 6:     5    70  70-74     AIS        AIS
 7:     6    71  70-74     PIS        PIS
 8:     7    72  70-74     PIS        PIS
 9:     8    73  70-74     PIS        PIS
10:     9    74  70-74     PIS        PIS
```

A patient who dies. The year of death is the last year recorded:

```
dcast(sim[id == died], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
   cycle   age agegrp         iabc   usual_care
   <int> <int> <char>       <char>       <char>
1:     0    65  65-69      NoEvent      NoEvent
2:     1    66  65-69      NoEvent      NoEvent
3:     2    67  65-69      NoEvent      NoEvent
4:     3    68  65-69      NoEvent      NoEvent
5:     4    69  65-69      NoEvent      NoEvent
6:     5    70  70-74      NoEvent      NoEvent
7:     6    71  70-74 CurrentDeath CurrentDeath
```

A patient who enters late, in cycle 7 (2020), and is followed until the end of the horizon:

```
dcast(sim[id == late], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
   cycle   age agegrp    iabc usual_care
   <int> <int> <char>  <char>     <char>
1:     7    65  65-69 NoEvent    NoEvent
2:     8    66  65-69 NoEvent    NoEvent
3:     9    67  65-69 NoEvent    NoEvent
```

A patient whose path differs between the arms. Both arms use the same random number for the same patient and year, so the paths stay together until, in one year, the same number lands on a different state in each arm's row:

```
dcast(sim[id == differs], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
    cycle   age agegrp    iabc usual_care
    <int> <int> <char>  <char>     <char>
 1:     0    71  70-74 NoEvent    NoEvent
 2:     1    72  70-74 NoEvent    NoEvent
 3:     2    73  70-74 NoEvent    NoEvent
 4:     3    74  70-74 NoEvent    NoEvent
 5:     4    75  75-79 NoEvent    NoEvent
 6:     5    76  75-79 NoEvent    NoEvent
 7:     6    77  75-79     AHS        AIS
 8:     7    78  75-79     PHS        PIS
 9:     8    79  75-79     PHS        PIS
10:     9    80  80-84     PHS        PIS
```

3.4 Their costs and QALYs, by hand

`tm_add_outcomes()` gives each patient-year its cost, intervention cost, QALY and discount factor:

```
out <- tm_add_outcomes(sim, costs, utilities, intervention, death, settings)
out[id == differs, .(arm, cycle, state, cost, intervention_cost, qaly, discount)]
```

```
          arm cycle   state  cost intervention_cost  qaly  discount
       <char> <int>  <char> <num>             <num> <num>     <num>
 1:       iabc     0 NoEvent  2000                 0  0.75 1.0000000
 2:       iabc     1 NoEvent  2050                50  0.75 0.9708738
 3:       iabc     2 NoEvent  2000                 0  0.75 0.9425959
 4:       iabc     3 NoEvent  2000                 0  0.75 0.9151417
 5:       iabc     4 NoEvent  2000                 0  0.75 0.8884870
 6:       iabc     5 NoEvent  2000                 0  0.75 0.8626088
 7:       iabc     6     AHS 20000                 0  0.50 0.8374843
 8:       iabc     7     PHS  5000                 0  0.60 0.8130915
 9:       iabc     8     PHS  5000                 0  0.60 0.7894092
10:       iabc     9     PHS  5000                 0  0.60 0.7664167
11: usual_care     0 NoEvent  2000                 0  0.75 1.0000000
12: usual_care     1 NoEvent  2000                 0  0.75 0.9708738
13: usual_care     2 NoEvent  2000                 0  0.75 0.9425959
14: usual_care     3 NoEvent  2000                 0  0.75 0.9151417
15: usual_care     4 NoEvent  2000                 0  0.75 0.8884870
16: usual_care     5 NoEvent  2000                 0  0.75 0.8626088
17: usual_care     6     AIS 15000                 0  0.55 0.8374843
18: usual_care     7     PIS  4000                 0  0.65 0.8130915
19: usual_care     8     PIS  4000                 0  0.65 0.7894092
20: usual_care     9     PIS  4000                 0  0.65 0.7664167
          arm cycle   state  cost intervention_cost  qaly  discount
       <char> <int>  <char> <num>             <num> <num>     <num>
```

The same numbers worked out by hand for the five patients, from the input tables:

- **Values:** in a patient's entry year both arms take the usual-care cost and utility; from the next year, each arm's own.
- **Intervention cost:** iABC patients pay the intervention cost of that year since entry, for each year after entry that they are alive.
- **Discount factor:** a year in cycle t is discounted by 1 / (1 + rate)^t.

```
costs_long <- melt(costs, id.vars = "state", variable.name = "arm", value.name = "state_cost",
 ↪  variable.factor = FALSE)
utilities_long <- melt(utilities, id.vars = "state", variable.name = "arm", value.name =
 ↪  "utility",
                       variable.factor = FALSE)
followed <- c(event_free, stroke, died, late, differs)
by_hand <- out[id %in% followed, .(id, arm, cycle, state, cost, intervention_cost, qaly,
 ↪  discount)]
by_hand[, years_after_entry := cycle - min(cycle), by = id]
by_hand[, value_arm := fifelse(years_after_entry == 0, "usual_care", arm)]
by_hand[costs_long, on = .(state, value_arm = arm), state_cost := i.state_cost]
by_hand[utilities_long, on = .(state, value_arm = arm), utility := i.utility]
by_hand[, programme := fifelse(arm == "iabc" & years_after_entry > 0 & !state %in% death,
                               intervention$cost[match(years_after_entry, intervention$year)],
 ↪  0)]
by_hand[, `:=`(cost_by_hand = state_cost + programme, discount_by_hand = 1 / (1 +
 ↪  settings$discount_rate)^cycle)]
by_hand[id == differs, .(arm, cycle, state, state_cost, programme, cost_by_hand, cost)]
```

```
          arm cycle   state state_cost programme cost_by_hand  cost
       <char> <int>  <char>      <num>     <num>        <num> <num>
 1:       iabc     0 NoEvent       2000         0         2000  2000
 2:       iabc     1 NoEvent       2000        50         2050  2050
 3:       iabc     2 NoEvent       2000         0         2000  2000
 4:       iabc     3 NoEvent       2000         0         2000  2000
 5:       iabc     4 NoEvent       2000         0         2000  2000
 6:       iabc     5 NoEvent       2000         0         2000  2000
 7:       iabc     6     AHS      20000         0        20000 20000
 8:       iabc     7     PHS       5000         0         5000  5000
 9:       iabc     8     PHS       5000         0         5000  5000
10:       iabc     9     PHS       5000         0         5000  5000
11: usual_care     0 NoEvent       2000         0         2000  2000
12: usual_care     1 NoEvent       2000         0         2000  2000
13: usual_care     2 NoEvent       2000         0         2000  2000
14: usual_care     3 NoEvent       2000         0         2000  2000
15: usual_care     4 NoEvent       2000         0         2000  2000
16: usual_care     5 NoEvent       2000         0         2000  2000
17: usual_care     6     AIS      15000         0        15000 15000
18: usual_care     7     PIS       4000         0         4000  4000
19: usual_care     8     PIS       4000         0         4000  4000
20: usual_care     9     PIS       4000         0         4000  4000
          arm cycle   state state_cost programme cost_by_hand  cost
       <char> <int>  <char>      <num>     <num>        <num> <num>
```

```
by_hand[id == differs, .(arm, cycle, state, utility, qaly, discount_by_hand =
 ↪  round(discount_by_hand, 4),
                          discount = round(discount, 4))]
```

```
          arm cycle   state utility  qaly discount_by_hand discount
       <char> <int>  <char>   <num> <num>           <num>    <num>
 1:       iabc     0 NoEvent    0.75  0.75          1.0000   1.0000
 2:       iabc     1 NoEvent    0.75  0.75          0.9709   0.9709
 3:       iabc     2 NoEvent    0.75  0.75          0.9426   0.9426
```

```
 4:       iabc     3 NoEvent    0.75  0.75           0.9151   0.9151
 5:       iabc     4 NoEvent    0.75  0.75           0.8885   0.8885
 6:       iabc     5 NoEvent    0.75  0.75           0.8626   0.8626
 7:       iabc     6     AHS    0.50  0.50           0.8375   0.8375
 8:       iabc     7     PHS    0.60  0.60           0.8131   0.8131
 9:       iabc     8     PHS    0.60  0.60           0.7894   0.7894
10:       iabc     9     PHS    0.60  0.60           0.7664   0.7664
11: usual_care     0 NoEvent    0.75  0.75           1.0000   1.0000
12: usual_care     1 NoEvent    0.75  0.75           0.9709   0.9709
13: usual_care     2 NoEvent    0.75  0.75           0.9426   0.9426
14: usual_care     3 NoEvent    0.75  0.75           0.9151   0.9151
15: usual_care     4 NoEvent    0.75  0.75           0.8885   0.8885
16: usual_care     5 NoEvent    0.75  0.75           0.8626   0.8626
17: usual_care     6     AIS    0.55  0.55           0.8375   0.8375
18: usual_care     7     PIS    0.65  0.65           0.8131   0.8131
19: usual_care     8     PIS    0.65  0.65           0.7894   0.7894
20: usual_care     9     PIS    0.65  0.65           0.7664   0.7664
           arm cycle   state utility  qaly discount_by_hand discount
        <char> <int>  <char>   <num> <num>            <num>    <num>
```

Every year of the five patients, compared with the package's numbers:

```
by_hand[, .(years = .N, same_cost = all(cost == cost_by_hand), same_intervention_cost =
 ↪  all(intervention_cost == programme),
            same_qaly = all(qaly == utility), same_discount = all(abs(discount -
             ↪  discount_by_hand) < 1e-12)), by = id]
```

```
      id years same_cost same_intervention_cost same_qaly same_discount
   <int> <int>    <lgcl>                 <lgcl>    <lgcl>        <lgcl>
1:     1    20      TRUE                   TRUE      TRUE          TRUE
2:     2    20      TRUE                   TRUE      TRUE          TRUE
3:     4    14      TRUE                   TRUE      TRUE          TRUE
4:   476    20      TRUE                   TRUE      TRUE          TRUE
5:  4179     6      TRUE                   TRUE      TRUE          TRUE
```

Each patient's discounted total in each arm, which is what the arm totals add up:

```
dcast(by_hand[, .(cost = sum(cost_by_hand * discount_by_hand), qalys = sum(utility *
 ↪  discount_by_hand)), by = .(id, arm)],
      id ~ arm, value.var = c("cost", "qalys"))
```

```
Key: <id>
      id cost_iabc cost_usual_care qalys_iabc qalys_usual_care
   <int>     <num>           <num>      <num>            <num>
1:     1 17620.762       17572.218   6.589582         6.589582
2:     2 35247.479       35198.936   6.096420         6.096420
3:     4 13720.411       13671.867   4.184780         4.184780
4:   476 39802.231       33197.348   6.024873         6.185193
5:  4179  4777.305        4737.835   1.776688         1.776688
```

4 How the model behaves over time

4.1 States over time

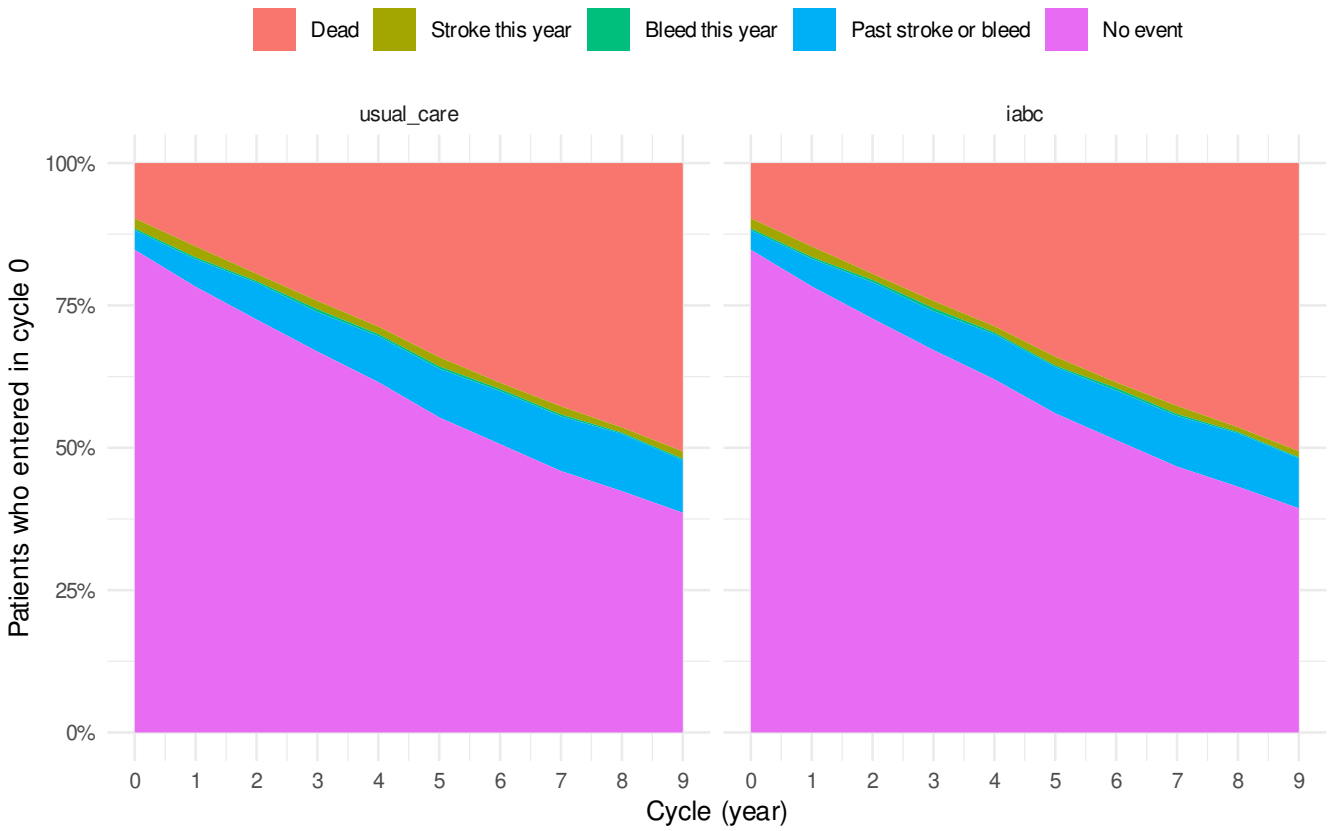
To see the cohort move through the states, the next plots follow the 2,664 patients who enter in cycle 0, so later entrants don't blur the picture. The 19 states are grouped into five:

```
state_groups <- data.table(state = states)
state_groups[, group := fcase(state == "NoEvent", "No event",
                              state %in% death, "Dead",
                              state %in% c("AIS", "AHS", "AIS_AB", "AHS_AB", "PB_AIS",
 ↪  "PB_AHS"), "Stroke this year",
                              state %in% c("AB", "PS_AB"), "Bleed this year",
                              default = "Past stroke or bleed")]
state_groups[, .(states = paste(state, collapse = ", ")), by = group]
```

```
                group
               <char>
1:            No event
2:     Bleed this year
3:    Stroke this year
4: Past stroke or bleed
5:                Dead
                                                                  states
                                                                  <char>
1:                                                                NoEvent
2:                                                              AB, PS_AB
3:                                  AHS, AIS, AHS_AB, AIS_AB, PB_AHS, PB_AIS
4:                                                    PB, PHS, PIS, PS_PB
5: CurrentDeath, AB_Death, AHS_Death, AIS_Death, AHS_AB_Death, AIS_AB_Death
```

Each year, the share of these patients in each group. A patient who has died stays counted as dead in later years. The patients already dead in cycle 0 are those whose entry state is a death state: they died in their first AF year. Both arms start from the same entry states, so they are the same in both arms:

```
first_entrants <- cohort[entry_cycle == 0, id]
yearly <- sim[id %in% first_entrants][state_groups, on = "state", nomatch = NULL][, .N, by =
 ↪  .(arm, cycle, group)]
# Years with nobody in a group count 0.
yearly <- yearly[CJ(arm = c("usual_care", "iabc"), cycle = 0:(settings$horizon - 1), group =
 ↪  unique(state_groups$group)),
                 on = .(arm, cycle, group)][is.na(N), N := 0L]
alive <- yearly[group != "Dead"]
dead <- yearly[group == "Dead"][order(cycle), .(cycle, N = cumsum(N), group = "Dead"), by =
 ↪  arm]
over_time <- rbind(alive, dead)[, share := N / length(first_entrants)]
over_time[, `:=`(group = factor(group, levels = c("Dead", "Stroke this year", "Bleed this
 ↪  year", "Past stroke or bleed",
                                                   "No event")), arm = factor(arm, levels =
                                                    ↪  c("usual_care", "iabc")))]
ggplot(over_time, aes(cycle, share, fill = group)) +
  geom_area(stat = "identity") + facet_wrap(~arm) +
  scale_x_continuous(breaks = 0:9) + scale_y_continuous(labels = scales::percent) +
  labs(x = "Cycle (year)", y = "Patients who entered in cycle 0", fill = NULL) +
  theme_minimal(base_size = 10) + theme(legend.position = "top")
```



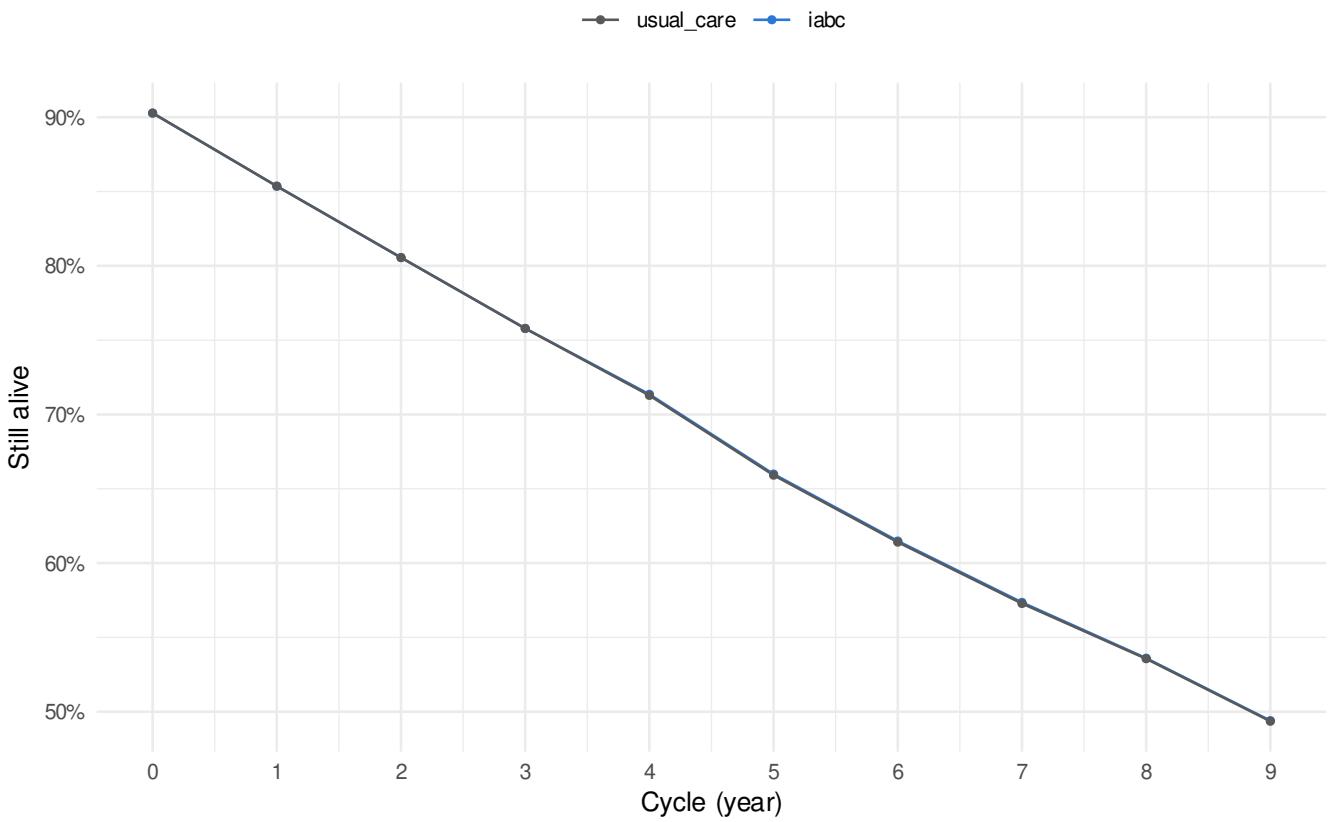

```
dcast(over_time[cycle %in% c(0, 5, 9)], arm + group ~ paste("cycle", cycle), value.var =
 ↪  "share")
```

```
Key: <arm, group>
          arm                group     cycle 0     cycle 5     cycle 9
       <fctr>               <fctr>       <num>       <num>       <num>
 1: usual_care                 Dead 0.097222222 0.340840841 0.506381381
 2: usual_care     Stroke this year 0.017642643 0.016141141 0.012762763
 3: usual_care      Bleed this year 0.003378378 0.004129129 0.002627628
 4: usual_care Past stroke or bleed 0.034159159 0.085585586 0.092342342
 5: usual_care             No event 0.847597598 0.553303303 0.385885886
 6:       iabc                 Dead 0.097222222 0.340090090 0.506006006
 7:       iabc     Stroke this year 0.017642643 0.015390390 0.010510511
 8:       iabc      Bleed this year 0.003378378 0.002627628 0.001876877
 9:       iabc Past stroke or bleed 0.034159159 0.081456456 0.087837838
10:       iabc             No event 0.847597598 0.560435435 0.393768769
```

4.2 Survival

The share of these patients still alive at the end of each year, and the deaths each year in each arm. The two lines lie almost on top of each other: in this run, iABC changes the number of deaths among these 2,664 patients by only a few, as the table shows:

```
survival <- dead[, .(arm, cycle, alive = 1 - N / length(first_entrants))]
ggplot(survival, aes(cycle, alive, colour = arm)) +
  geom_line() + geom_point(size = 1) + scale_colour_manual(values = arm_colours, breaks =
   ↪  names(arm_colours)) +
  scale_x_continuous(breaks = 0:9) + scale_y_continuous(labels = scales::percent) +
  labs(x = "Cycle (year)", y = "Still alive", colour = NULL) +
  theme_minimal(base_size = 10) + theme(legend.position = "top")
```




```
deaths_by_year <- dcast(yearly[group == "Dead"], cycle ~ arm, value.var = "N")
deaths_by_year[, `:=`(difference = iabc - usual_care, cumulative_difference = cumsum(iabc -
 ↪  usual_care))]
deaths_by_year
```

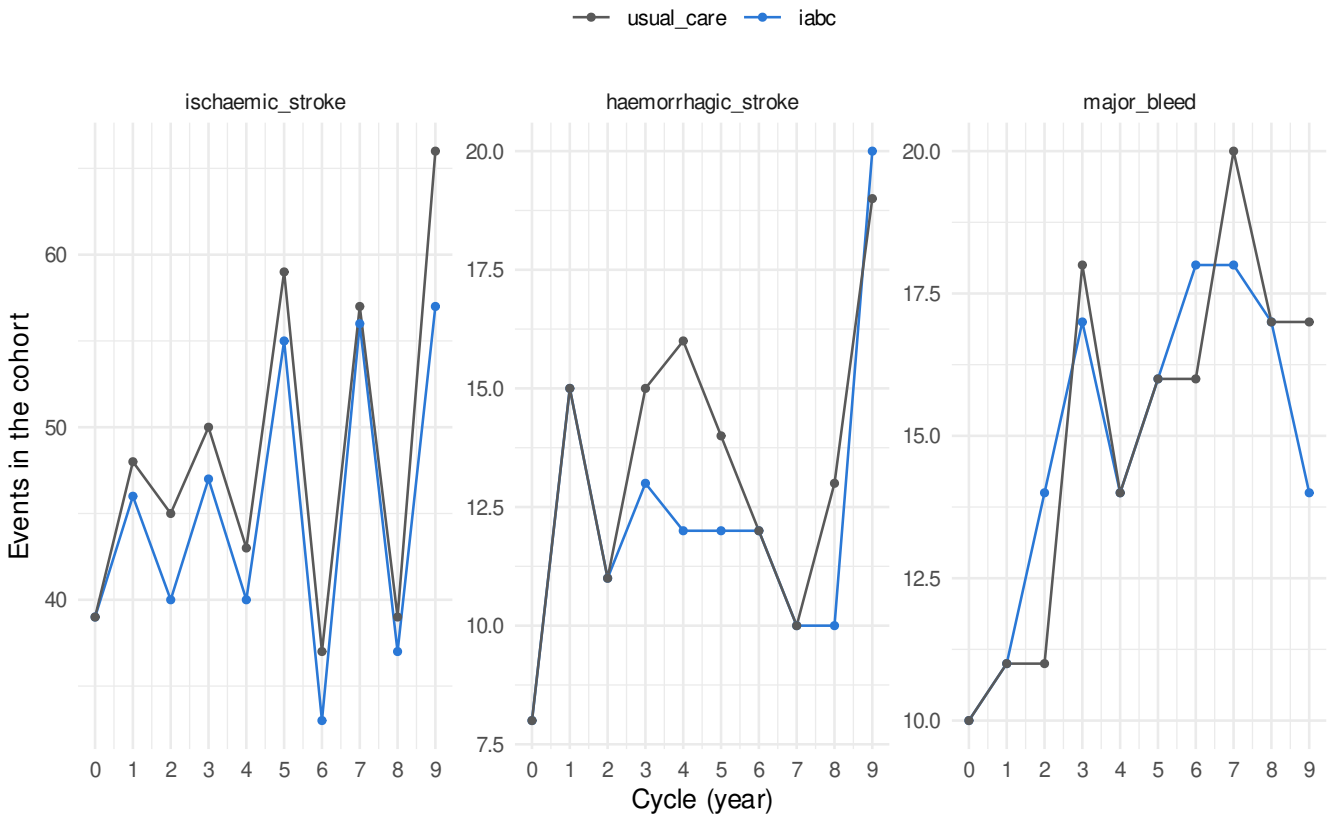
```
Key: <cycle>
    cycle  iabc usual_care difference cumulative_difference
    <int> <int>      <int>      <int>                 <int>
 1:     0   259        259          0                     0
 2:     1   131        131          0                     0
 3:     2   128        128          0                     0
 4:     3   127        127          0                     0
 5:     4   118        120         -2                    -2
 6:     5   143        143          0                    -2
 7:     6   120        120          0                    -2
 8:     7   110        110          0                    -2
 9:     8   100         99          1                    -1
10:     9   112        112          0                    -1
```

4.3 Strokes and bleeds per year

The events counted in the results (`events` in `settings.yaml`), each year, for the whole cohort. Each year a patient spends in one of an event's states counts as one event:

```
event_states <- setnames(as.data.table(stack(lapply(settings$events, unlist))), c("state",
 ↪  "event"))
events_by_year <- sim[event_states, on = "state", allow.cartesian = TRUE, nomatch = NULL][, .N,
 ↪  by = .(arm, cycle, event)]
ggplot(events_by_year, aes(cycle, N, colour = arm)) +
  geom_line() + geom_point(size = 1) + facet_wrap(~event, scales = "free_y") +
   ↪  scale_colour_manual(values = arm_colours, breaks = names(arm_colours)) +
```

```
  scale_x_continuous(breaks = 0:9) +
  labs(x = "Cycle (year)", y = "Events in the cohort", colour = NULL) +
  theme_minimal(base_size = 10) + theme(legend.position = "top")
```




```
dcast(events_by_year, event ~ arm, value.var = "N", fun.aggregate = sum)[, difference := iabc
 ↪  - usual_care][]
```

```
Key: <event>
                event  iabc usual_care difference
               <fctr> <int>      <int>      <int>
1:    ischaemic_stroke   450        483        -33
2: haemorrhagic_stroke   123        133        -10
3:         major_bleed   149        150         -1
```

4.4 Costs and QALYs building up

The discounted cost and QALYs per patient, added up year by year, for the whole cohort. The cost is split into the intervention cost and the cost of care (the state costs):

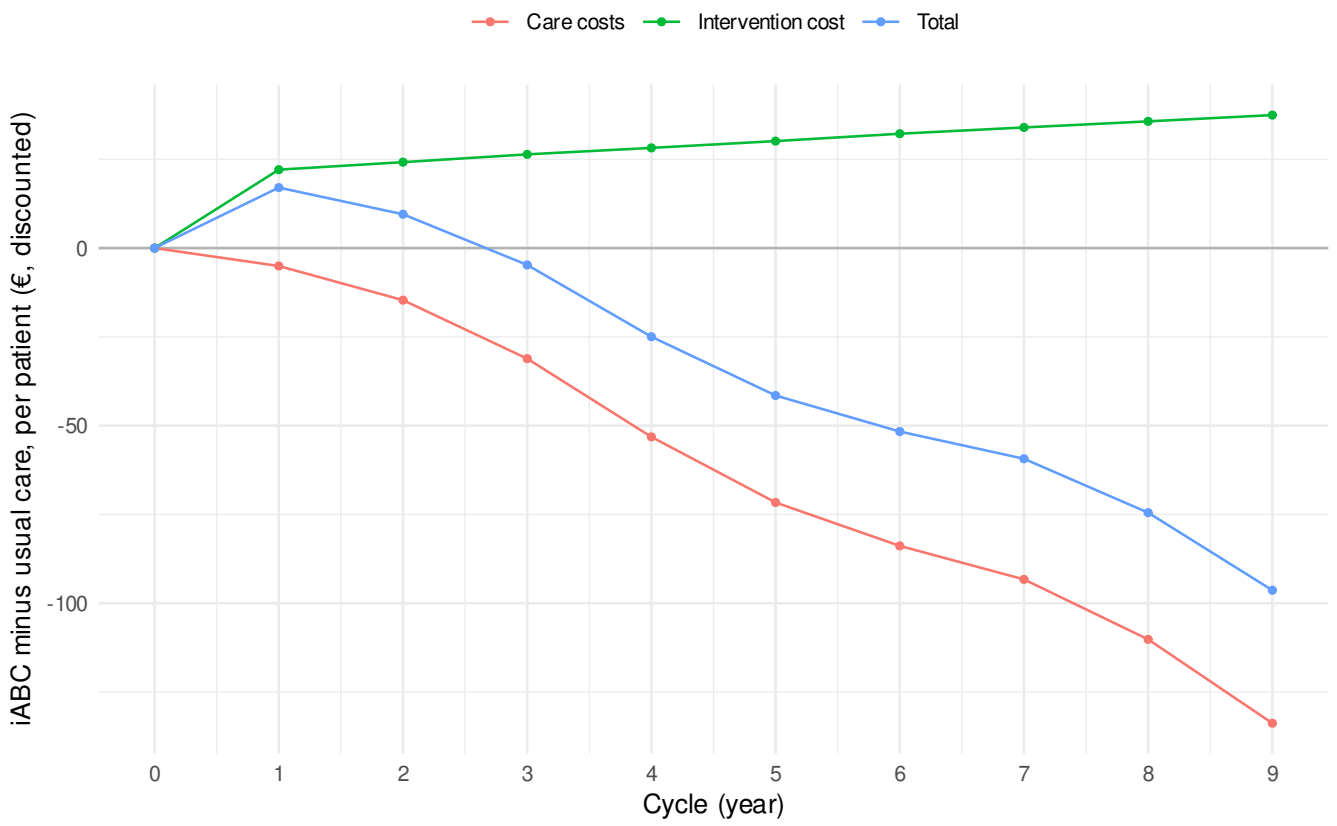
```
n_patients <- uniqueN(out$id)
by_year <- out[, .(care_cost = sum((cost - intervention_cost) * discount), intervention_cost =
 ↪  sum(intervention_cost * discount),
                   qalys = sum(qaly_disc)), by = .(arm, cycle)]
setorder(by_year, arm, cycle)
by_year[, `:=`(care_cost = cumsum(care_cost) / n_patients, intervention_cost =
 ↪  cumsum(intervention_cost) / n_patients,
               qalys = cumsum(qalys) / n_patients), by = arm]
difference <- dcast(by_year, cycle ~ arm, value.var = c("care_cost", "intervention_cost",
 ↪  "qalys"))
```

```
difference <- difference[, .(cycle, care_cost = care_cost_iabc - care_cost_usual_care,
                             intervention_cost = intervention_cost_iabc -
                              ↪  intervention_cost_usual_care,
                             total_cost = care_cost_iabc + intervention_cost_iabc -
                              ↪  care_cost_usual_care - intervention_cost_usual_care,
                             qalys = qalys_iabc - qalys_usual_care)]
difference
```

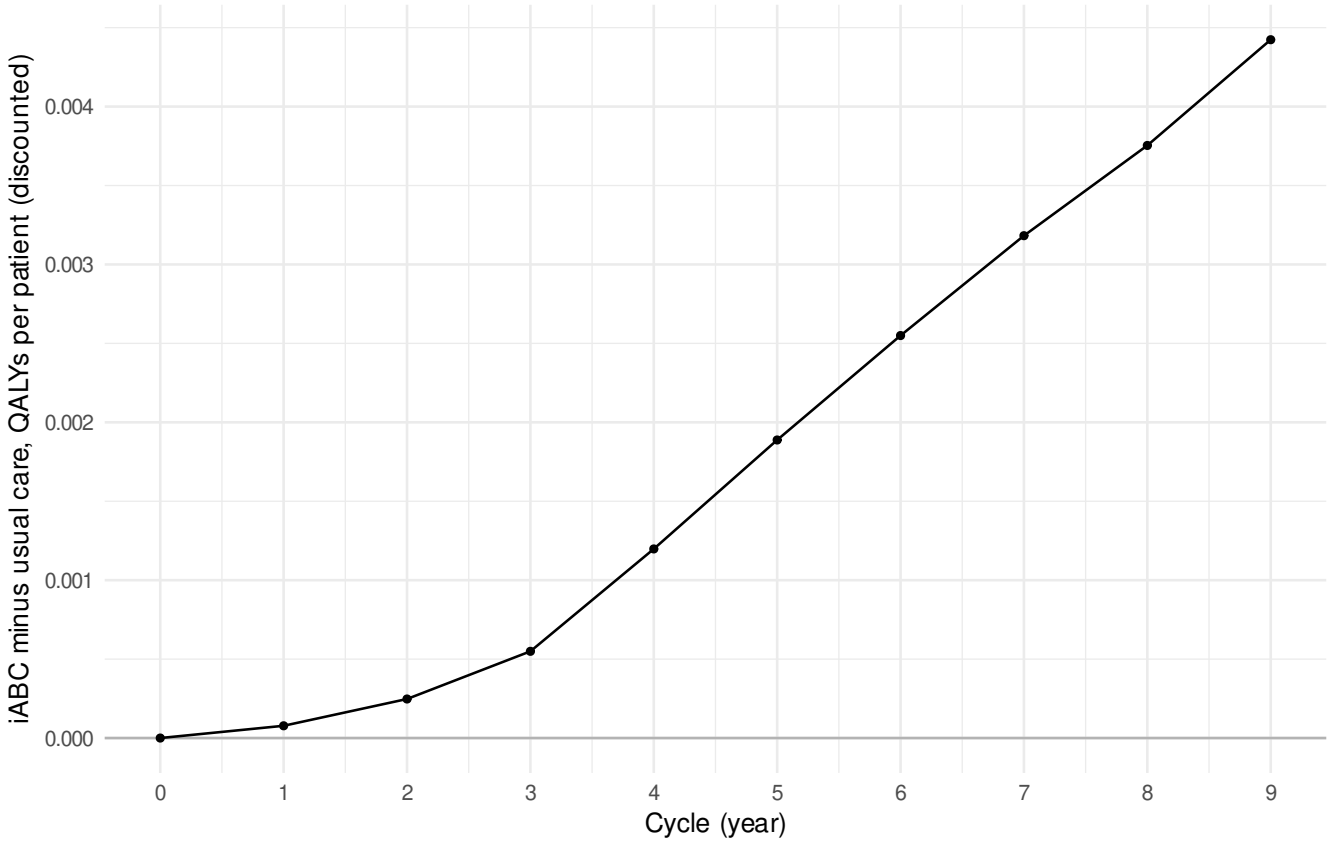
```
Key: <cycle>
    cycle   care_cost intervention_cost total_cost        qalys
    <int>       <num>             <num>      <num>        <num>
 1:     0    0.000000           0.00000   0.000000 0.0000000000
 2:     1   -5.048544          22.07767  17.029126 0.0000776699
 3:     2  -14.663022          24.18908   9.526063 0.0002473372
 4:     3  -31.135572          26.38542  -4.750147 0.0005493339
 5:     4  -53.170051          28.20682 -24.963227 0.0011979295
 6:     5  -71.629879          30.12181 -41.508064 0.0018880165
 7:     6  -83.857149          32.19878 -51.658373 0.0025496290
 8:     7  -93.289010          33.96318 -59.325826 0.0031822142
 9:     8 -110.182368          35.67620 -74.506166 0.0037537465
10:     9 -133.788003          37.44662 -96.341378 0.0044235948
```

iABC minus usual care, added up year by year. The intervention cost is paid in the first year after each patient's entry, so it rises most in cycle 1 and then a little each year as later entrants reach their first year. The care costs saved through avoided strokes and bleeds build up over the years:

```
cost_parts <- melt(difference, id.vars = "cycle", measure.vars = c("care_cost",
 ↪  "intervention_cost", "total_cost"))
cost_parts[, variable := factor(variable, labels = c("Care costs", "Intervention cost",
 ↪  "Total"))]
ggplot(cost_parts, aes(cycle, value, colour = variable)) +
  geom_hline(yintercept = 0, colour = "grey70") + geom_line() + geom_point(size = 1) +
  scale_x_continuous(breaks = 0:9) +
  labs(x = "Cycle (year)", y = "iABC minus usual care, per patient (€, discounted)", colour =
   ↪  NULL) +
  theme_minimal(base_size = 10) + theme(legend.position = "top")
```




```
ggplot(difference, aes(cycle, qalys)) +
  geom_hline(yintercept = 0, colour = "grey70") + geom_line() + geom_point(size = 1) +
  scale_x_continuous(breaks = 0:9) +
  labs(x = "Cycle (year)", y = "iABC minus usual care, QALYs per patient (discounted)") +
  theme_minimal(base_size = 10)
```



4.5 The run's summary and comparison

`tm_summarise()` adds up each arm, and `tm_compare()` takes iABC minus usual care:

```
one_run <- tm_summarise(out, death, settings)
one_run
```

```
         arm patients cost_total qaly_total intervention_cost_total
      <char>    <int>      <num>      <num>                   <num>
1:       iabc     5000   65618011   17969.23                187233.1
2: usual_care     5000   66099718   17947.12                     0.0
   cost_per_patient qaly_per_patient intervention_cost_per_patient ischaemic_stroke
              <num>            <num>                         <num>            <int>
1:         13123.60         3.593847                      37.44662              450
2:         13219.94         3.589423                       0.00000              483
   haemorrhagic_stroke major_bleed deaths
                 <int>       <int>  <int>
1:                 123         149   1845
2:                 133         150   1846
```

```
compared <- tm_compare(copy(one_run)[, `:=`(run = 1L, seed = seed)])
compared$overall
```

```
   delta_cost delta_intervention_cost  delta_qaly      icer quadrant
        <num>                   <num>       <num>     <num>   <char>
1:  -96.34138                37.44662 0.004423595 -21778.98 dominant
   delta_ischaemic_stroke delta_haemorrhagic_stroke delta_major_bleed delta_deaths
                    <num>                     <num>             <num>        <num>
1:                    -33                       -10                -1           -1
```

The incremental cost and QALYs are the last row of the year-by-year differences above:

```
difference[.N, .(total_cost, qalys)]
```

```
   total_cost       qalys
        <num>       <num>
1:  -96.34138 0.004423595
```

4.6 Subgroups

The discounted cost and QALYs of each patient, and the difference between the arms, by sex, by age group at entry and by cluster:

```
per_patient <- out[, .(cost = sum(cost_disc), qalys = sum(qaly_disc)), by = .(id, arm)][cohort,
 ↪  on = "id"]
per_patient <- dcast(per_patient, id + sex + agegrp + mm_cluster ~ arm, value.var = c("cost",
 ↪  "qalys"))
per_patient[, `:=`(delta_cost = cost_iabc - cost_usual_care, delta_qalys = qalys_iabc -
 ↪  qalys_usual_care)]
per_patient[, .(patients = .N, qalys_usual_care = mean(qalys_usual_care), delta_cost =
 ↪  mean(delta_cost),
                delta_qalys = mean(delta_qalys)), keyby = sex]
```

```
Key: <sex>
      sex patients qalys_usual_care delta_cost delta_qalys
   <char>    <int>            <num>      <num>       <num>
1:    men     2052         3.565699  -55.28531 0.005192154
2:  women     2948         3.605936 -124.91908 0.003888627
```

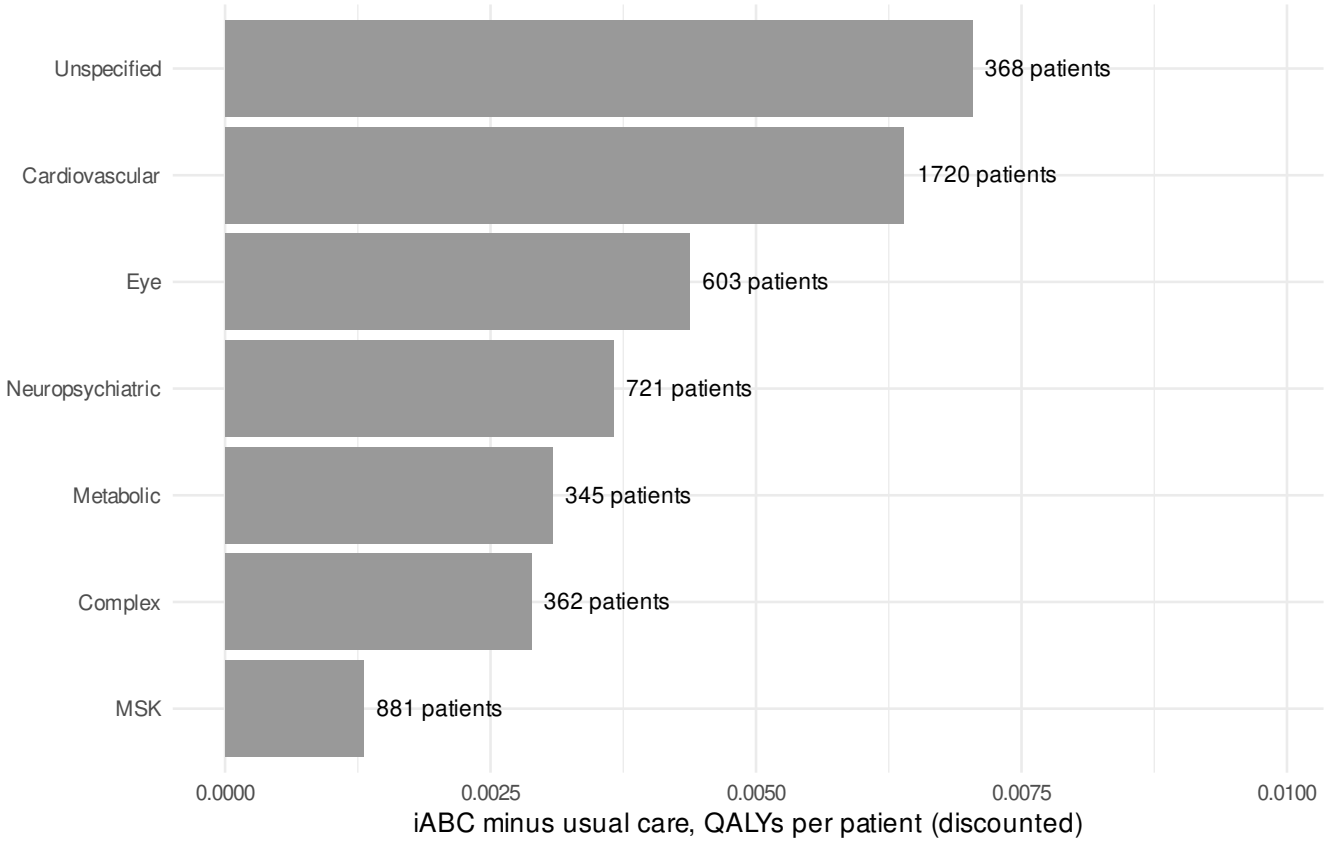
```
per_patient[, .(patients = .N, qalys_usual_care = mean(qalys_usual_care), delta_cost =
 ↪  mean(delta_cost),
                delta_qalys = mean(delta_qalys)), keyby = agegrp]
```

```
Key: <agegrp>
   agegrp patients qalys_usual_care delta_cost delta_qalys
   <char>    <int>            <num>      <num>       <num>
1:  65-69     1182         3.621257  -64.83610 0.002581692
2:  70-74      914         4.173160  -91.00585 0.003174663
3:  75-79     1010         4.035495 -125.91163 0.004061984
4:  80-84      935         3.458156 -111.88974 0.007025705
5:  85-89      630         2.874727  -94.20818 0.007203156
6:  90-94      267         2.274542 -121.31857 0.003577324
7:  95-99       62         2.014755   26.44476 0.000000000
```

```
by_cluster <- per_patient[, .(patients = .N, qalys_usual_care = mean(qalys_usual_care),
 ↪  delta_cost = mean(delta_cost),
                               delta_qalys = mean(delta_qalys)), keyby = mm_cluster]
by_cluster
```

```
Key: <mm_cluster>
         mm_cluster patients qalys_usual_care delta_cost delta_qalys
             <char>    <int>            <num>      <num>       <num>
1:   Cardiovascular     1720         3.633256  -78.93994 0.006389477
2:          Complex      362         2.781613 -130.78397 0.002882857
3:              Eye      603         3.686695 -130.65498 0.004377813
4:              MSK      881         3.958040  -22.70521 0.001308159
5:        Metabolic      345         3.760879 -135.70648 0.003085344
6: Neuropsychiatric      721         3.182828 -104.69996 0.003658971
7:      Unspecified      368         3.773203 -210.57268 0.007036981
```

```
ggplot(by_cluster, aes(reorder(mm_cluster, delta_qalys), delta_qalys)) +
  geom_col(fill = "grey60") + geom_text(aes(label = paste(patients, "patients")), hjust = -0.1,
   ↪  size = 3) +
  coord_flip() + expand_limits(y = max(by_cluster$delta_qalys) * 1.4) +
  labs(x = NULL, y = "iABC minus usual care, QALYs per patient (discounted)") +
  theme_minimal(base_size = 10)
```



These come from one run. In small groups, a handful of avoided events moves the average a lot, so subgroup results from a single run vary more than the overall result.

5 The full model: `tm_run()`

`tm_run()` does everything above `n_runs` times. Run r uses seed `seed + r - 1`, so its first run is the run followed in this tutorial. It returns the mean over runs and writes the results to `outputs/`:

```
results <- tm_run()
results
```

```
<tm_results> mean per arm over 100 runs (discounted costs and QALYs)

         arm patients cost_total qaly_total intervention_cost_total
      <char>    <num>      <num>      <num>                   <num>
1:       iabc     5000   65493591   18004.50                187494.6
2: usual_care     5000   66035435   17980.58                     0.0
   cost_per_patient qaly_per_patient intervention_cost_per_patient ischaemic_stroke
              <num>            <num>                         <num>            <num>
1:         13098.72            3.601                        37.499           429.23
2:         13207.09            3.596                         0.000           461.93
```

```
   haemorrhagic_stroke major_bleed  deaths
                 <num>       <num>   <num>
1:              121.54      147.47 1850.85
2:              129.50      160.45 1853.00

iABC minus usual care (mean over runs, per patient):
  Incremental cost:   €-108.37 (intervention cost: €37.5)
  Incremental QALYs:  0.00478
  ICER:               iABC dominant (cheaper, more QALYs)
  Difference in events (whole cohort): ischaemic_stroke -32.7, haemorrhagic_stroke -8.0, major_bleed
   ↪  -13.0, deaths -2.1

Written to outputs/: summary.csv, summary_by_run.csv, comparison.csv, comparison_by_run.csv, ce_plane.png
```

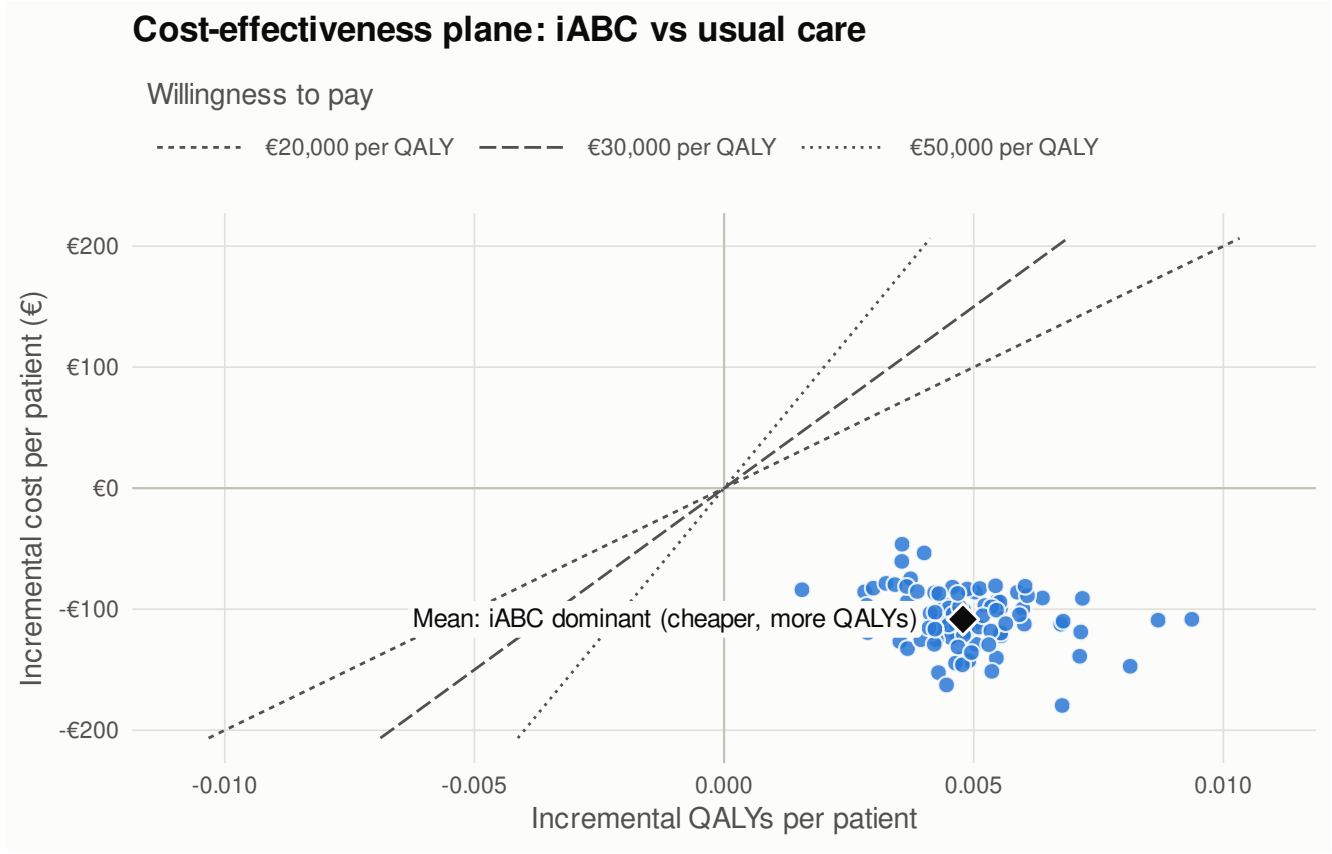
The first run of `tm_run()` and the run followed above, side by side:

```
rbind(cbind(source = "step by step", one_run[, .(arm, cost_per_patient, qaly_per_patient,
 ↪  deaths)]),
      cbind(source = "tm_run(), run 1", results$by_run[run == 1, .(arm, cost_per_patient,
       ↪  qaly_per_patient, deaths)]))
```

```
            source        arm cost_per_patient qaly_per_patient deaths
            <char>     <char>           <num>            <num>  <int>
1:    step by step       iabc        13123.60         3.593847   1845
2:    step by step usual_care        13219.94         3.589423   1846
3: tm_run(), run 1       iabc        13123.60         3.593847   1845
4: tm_run(), run 1 usual_care        13219.94         3.589423   1846
```

Each point on the cost-effectiveness plane is one run; the diamond is the mean across runs:

```
plot(results)
```



```r
list.files("outputs")
```

```
[1] "ce_plane.png"          "comparison_by_run.csv" "comparison.csv"
[4] "summary_by_run.csv"    "summary.csv"
```